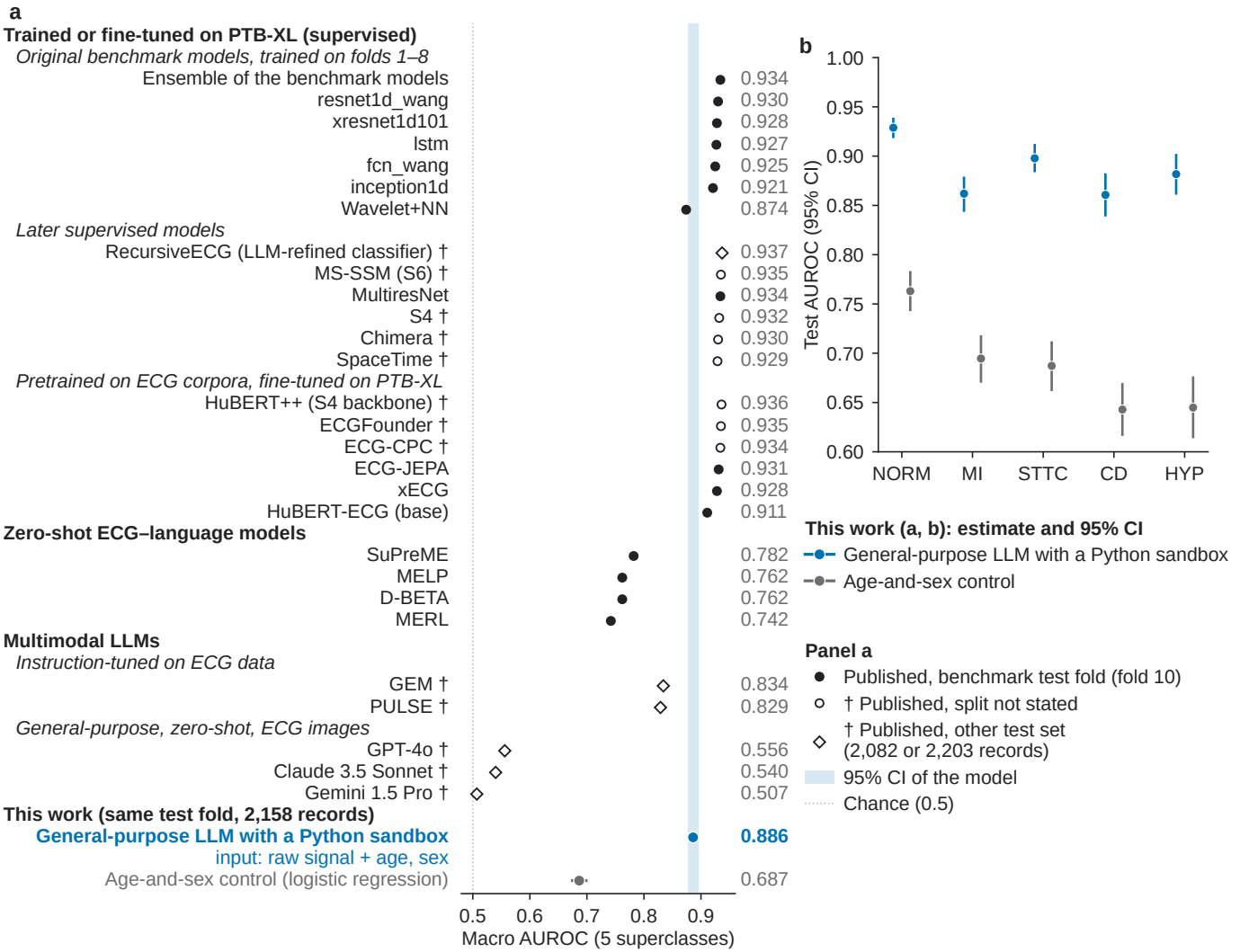
a
Trained or fine-tuned on PTB-XL (supervised)
Original benchmark models, trained on folds 1–8
Ensemble of the benchmark models 0.934
resnet1d_wang 0.930
xresnet1d101 0.928
lstm 0.927
fcn_wang 0.925
inception1d 0.921
Wavelet+NN 0.874
Later supervised models
RecursiveECG (LLM-refined classifier) † 0.937
MS-SSM (S6) † 0.935
MultiresNet 0.934
S4 † 0.932
Chimera † 0.930
SpaceTime † 0.929
Pretrained on ECG corpora, fine-tuned on PTB-XL
HuBERT++ (S4 backbone) † 0.936
ECGFounder † 0.935
ECG-CPC † 0.934
ECG-JEPA 0.931
xECG 0.928
HuBERT-ECG (base) 0.911
Zero-shot ECG–language models
SuPreME 0.782
MELP 0.762
D-BETA 0.762
MERL 0.742
Multimodal LLMs
Instruction-tuned on ECG data
GEM † 0.834
PULSE † 0.829
General-purpose, zero-shot, ECG images
GPT-4o † 0.556
Claude 3.5 Sonnet † 0.540
Gemini 1.5 Pro † 0.507
This work (same test fold, 2,158 records)
General-purpose LLM with a Python sandbox 0.886
input: raw signal + age, sex
Age-and-sex control (logistic regression) 0.687
0.5 0.6 0.7 0.8 0.9
Macro AUROC (5 superclasses)
b
1.00 0.95 0.90 0.85 0.80 0.75 0.70 0.65 0.60
Test AUROC (95% CI)
NORM MI STTC CD HYP
This work (a, b): estimate and 95% CI
General-purpose LLM with a Python sandbox
Age-and-sex control
Panel a
Published, benchmark test fold (fold 10)
† Published, split not stated
† Published, other test set (2,082 or 2,203 records)
95% CI of the model
Chance (0.5)